Generative Adversarial Fourier Neural Operator for CFD

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Abstract

In this work, a Generative Adversarial Fourier Neural Operator (GA-FNO) is presented for surrogate modeling of 2D Kolmogorov turbulence. The architecture employs two specialized discriminators: one evaluating the statistical coherence of generated trajectories and one evaluating Navier-Stokes residuals between frames. Zero-shot generalization to extended spatial domains is demonstrated through domain decomposition with cosine blending. Additionally, super-resolution capabilities and an Inverse Boundary Method for object simulation within generated flow fields are explored as post-processing extensions.

In the document Mathematical and Experimental details are discussed also autoregressive limitations are presented, in particular the emergence of spurious attractors in long-horizon rollouts, providing insights for future architecture and training improvements.

Keywords: ¡keywords¿

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1 Introduction

The world of machine learning and numerical simulations are deeply tied, as machine learning methods can leverage simulation data to optimize and accelerate real life applications. Following this, we have approached the problem with a generative adversarial focus, looking for the probabilistic distribution that best replicates the non-linear dynamics of fluid flow.

Generative Adversarial Networks present a natural framework for this problem for three key reasons. First, they are the only generative framework that learns a distribution without explicit supervision — the generator does not need to know where the ground truth is, it only needs to fool the discriminator. Second, the adversarial nature is well suited for turbulence — two competing networks capture chaotic dynamics more naturally than deterministic models. Third, the discriminator acts as an implicit physical critic, learning what makes a trajectory physically plausible without being explicitly told.

Building on this, we extend the standard GAN framework with two specialized discriminators. The first evaluates the statistical coherence of the generated trajectories as a whole, while the second evaluates the Navier-Stokes residual between frames — ensuring that the generated flow respects the underlying physical equations at every timestep. The generator itself is implemented as a Neural Operator, which learns the solution operator of the flow dynamics directly in Fourier spectral space.

2 Mathematical Foundations

2.1 Navier-Stokes in Vorticity Form

The 2D incompressible Navier-Stokes equations are expressed in vorticity-stream function form. Let $\omega = \nabla \times u$ be the vorticity field. The governing equation is:

$$\frac{\partial \omega}{\partial t} + u\frac{\partial \omega}{\partial x} + v\frac{\partial \omega}{\partial y} = \nu \nabla^2 \omega + f \tag{1}$$

where $\nu = UL/Re$ is the kinematic viscosity, $f = \sin(Ky)$ is the Kolmogorov forcing, and (u, v) is the velocity field recovered from the stream function ψ via:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}, \quad \nabla^2 \psi = -\omega \tag{2}$$

The Navier-Stokes residual, used as input to the physical discriminator, is defined as:

$$R = \frac{\partial \omega}{\partial t} + u\frac{\partial \omega}{\partial x} + v\frac{\partial \omega}{\partial y} - \nu \nabla^2 \omega - f \tag{3}$$

If $R = 0$ the field is physically consistent; otherwise $R \neq 0$ quantifies the violation. Discretizing in time:

$$\omega^{n+1} = \omega^n + \Delta t \left(-\frac{\partial \psi}{\partial y}\frac{\partial \omega}{\partial x} + \frac{\partial \psi}{\partial x}\frac{\partial \omega}{\partial y} + \nu \nabla^2 \omega + f \right) \tag{4}$$

2.2 Spectral Solver

The stream function is recovered spectrally in three steps. First, the FFT maps the vorticity to Fourier space:

$$\hat{\omega}(k_x, k_y) = \sum_{x,y} \omega(x, y)\, e^{-i(k_x x + k_y y)} \tag{5}$$

Second, the Poisson equation $\nabla^2 \psi = -\omega$ becomes algebraic in Fourier space:

$$\hat{\psi}_{k_x,k_y} = \frac{\hat{\omega}_{k_x,k_y}}{k_x^2 + k_y^2} \tag{6}$$

Third, the physical stream function is recovered via the inverse FFT:

$$\psi(x, y) = \sum_{k_x,k_y} \hat{\psi}(k_x, k_y)\, e^{i(k_x x + k_y y)} \tag{7}$$

from which the velocity components are obtained as $u = \partial_y \psi$, $v = -\partial_x \psi$, computed spectrally as $\hat{u} = ik_y\hat{\psi}$, $\hat{v} = ik_x\hat{\psi}$, followed by IFFT.

2.3 Spectral Convolution

Given an input field $w(x, y)$, the spectral convolution layer first maps it to Fourier space:

$$\hat{w}(k_x, k_y) = \mathcal{F}(w) \tag{8}$$

A complex-valued linear transformation is then applied to the spectral coefficients. The output representation $\hat{h}$ is obtained by multiplying $\hat{w}$ with a trainable weight tensor:

$$\hat{h}_o(k_x, k_y) = \sum_{i=1}^{C_{\text{in}}} W_{io}(k_x, k_y)\, \hat{w}_i(k_x, k_y) \tag{9}$$

where C_{in} and C_{out} denote the number of input and output channels, respectively, and $W_{io}(k_x, k_y) \in \mathbb{C}$ are complex weights. These are initialized as:

$$W = \frac{1}{C_{\text{in}} C_{\text{out}}} \left(R_r + iR_i \right), \quad R_r, R_i \sim \mathcal{N}(0, 1) \tag{10}$$

In practice, this operation is restricted to a truncated set of Fourier modes, retaining only low-frequency components.

Finally, the output field in physical space is obtained via the inverse transform:

$$h(x, y) = \mathcal{F}^{-1}(\hat{h}) \tag{11}$$

This formulation corresponds to a global convolution operator, implemented efficiently through FFTs, where the kernel is learned directly in spectral space.

2.4 Wasserstein Distance and Gradient Penalty

By the Kantorovich-Rubinstein duality, the Wasserstein-1 distance between the real distribution p_r and the generated distribution p_g is:

$$W_1(p_r, p_g) = \sup_{\|\nabla D\|_2 \leq 1} \int D(x)\big(p_r(x) - p_g(x)\big)\, dx \tag{12}$$

which in expectation form reads $\mathbb{E}_{p_r}[D(x)] - \mathbb{E}_{p_g}[D(x)]$. The 1-Lipschitz constraint $|D(x) - D(G(z))| \leq \|x - G(z)\|$ is enforced softly via a gradient penalty. The discriminator loss is:

$$\mathcal{L}_D = \mathbb{E}[D(G(z))] - \mathbb{E}[D(x)] + \lambda\, \mathbb{E}_{\hat{x}}\Big[\big(\|\nabla D(\hat{x})\|_2 - 1\big)^2\Big] \tag{13}$$

where the interpolated samples are drawn as:

$$\hat{x} = \alpha\, x + (1 - \alpha)\, G(z), \quad \alpha \sim \mathcal{U}[0, 1] \tag{14}$$

2.5 Inverse Boundary Method

The Inverse Boundary Method is integrated into the vorticity–stream function formulation of the 2D incompressible Navier–Stokes equations to handle embedded obstacles within a periodic domain. The boundary condition is enforced at the level of the Poisson solve through a volume penalization approach. Given a predicted vorticity field ω, a free stream function ψ is first obtained from the Poisson equation:

$$\nabla^2 \psi - \chi(x, y)\, \frac{1}{\eta}\big(\psi - \psi_s\big) = -\omega \tag{15}$$

where $\chi(x, y)$ is a smooth indicator function of the obstacle and $\eta \ll 1$ is the penalization parameter. This enforces $\psi \to \psi_b$ inside the obstacle, approximating the boundary condition.

In practice, this is implemented iteratively. Starting from the free solution $\psi^{(0)}$, the stream function is updated using a relaxed Jacobi-like iteration. The residual at step k is defined as:

$$\Delta\psi^{(k)} = -\nabla^2 \psi^{(k)} + \frac{\chi}{\eta}\big(\psi^{(k)} - \psi_s\big) - \omega \tag{16}$$

and the update step is:

$$\psi^{(k+1)} = \psi^{(k)} + \beta\,\Delta\psi^{(k)} \tag{17}$$

where $\beta \in (0, 1]$ is a relaxation parameter. After convergence, the corrected stream function ψ_{corr} is used to recover the velocity field and the vorticity:

$$\omega^{(k)} = -\nabla^2\psi^{(k)} \tag{18}$$

This modified projection step is embedded within each time update of the Navier–Stokes dynamics, allowing the evolution of ω to respect obstacle-induced constraints without breaking the spectral structure of the solver.

2.6 Model Structure

2.6.1 Generator

The generator G_θ is implemented as a Fourier Neural Operator (FNO) that learns a mapping between consecutive vorticity fields. Given the current state $\omega^n \in \mathbb{R}^{H\times W}$ and a latent vector $z \sim \mathcal{N}(0, I)$, the input is first lifted to a higher-dimensional feature space:

$$h_{ij}^{(0)} = W_{\text{lift}} \cdot [\omega_{ij}^n,\, z] \tag{19}$$

A sequence of L FNO blocks is then applied. Each block consists of a spectral convolution combined with a local linear transformation and a nonlinearity:

$$h^{(\ell+1)} = \text{GELU}\left(\frac{\mathcal{K}^{(\ell)}(h^{(\ell)}) + W_{\text{conv}}^{(\ell)}\, h^{(\ell)} - \mu^{(\ell)}}{\sigma^{(\ell)}}\right), \quad \ell = 0, \ldots, L-1 \tag{20}$$

where each block has independent trainable parameters, including spectral weights, pointwise linear mappings, and normalization statistics. This stacked structure enables the model to learn progressively richer representations in Fourier space.

After the final block, the feature map $h^{(L)} \in \mathbb{R}^{C\times H\times W}$ is projected back to a scalar field:

$$h(\omega) = W_{\text{proj}} \cdot h^{(L)}, \quad h(\omega) \in \mathbb{R}^{H\times W} \tag{21}$$

Finally, a residual connection produces the next state:

$$\omega^{n+1} = \omega^n + h(\omega) \tag{22}$$

2.6.2 Discriminators

Two discriminators with identical architectures are employed, differing only in their input.

Statistical discriminator $D_{\mathbf{stat}}$. Receives a full generated trajectory

$$\mathcal{T} = \begin{pmatrix} \omega^0 \\ \omega^1 \\ \vdots \\ \omega^9 \end{pmatrix} \in \mathbb{R}^{10\times H\times W}$$

A temporal convolutional encoder produces $h \in \mathbb{R}^{64\times H\times W}$:

$$\mathcal{T} \longrightarrow h \in \mathbb{R}^{64\times H\times W} \tag{23}$$

The convolution across channels is given by:

$$h_k(x,y) = \sum_{i=1}^{64} W_{k,i}\, \omega_i(x,y) \tag{24}$$

The representation is then processed by FNO blocks, preserving the dimensionality:

$$h \longrightarrow h' \in \mathbb{R}^{64\times H\times W} \tag{25}$$

A spatial average is applied:

$$\bar{h}_k = \frac{1}{HW} \sum_{x,y} h'_k(x,y) \tag{26}$$

yielding $\bar{h} \in \mathbb{R}^{64}$, which is passed through two linear layers with GELU activation:

$$D_{\text{stat}}(\mathcal{T}) = L_{n_1} \cdot \text{GELU}(L_{n_2} \cdot \bar{h}) \tag{27}$$

Physical discriminator $D_{\mathbf{phys}}$. Receives the sequence of Navier-Stokes residuals

$$R = \begin{pmatrix} R^0 \\ R^1 \\ \vdots \\ R^{T-2} \end{pmatrix} \in \mathbb{R}^{(T-1)\times H\times W}$$

A temporal convolutional encoder produces $h \in \mathbb{R}^{64\times H\times W}$:

$$R \longrightarrow h \in \mathbb{R}^{64\times H\times W} \tag{28}$$

The convolution across channels is given by:

$$h_k(x,y) = \sum_{i=1}^{64} W_{k,i}\, R_i(x,y) \tag{29}$$

The representation is then processed by FNO blocks, preserving the dimensionality:

$$h \longrightarrow h' \in \mathbb{R}^{64\times H\times W} \tag{30}$$

A spatial average is applied:

$$\bar{h}_k = \frac{1}{HW}\sum_{x,y} h'_k(x,y) \tag{31}$$

yielding $\bar{h} \in \mathbb{R}^{64}$, which is passed through two linear layers with GELU activation:

$$D_{\text{phys}}(R) = L_{n_1} \cdot \text{GELU}(L_{n_2} \cdot \bar{h}) \tag{32}$$

2.7 Objective Function

The full training objective couples both discriminators. Each discriminator is updated independently with its own gradient penalty:

$$\mathcal{L}_{D_{\text{stat}}} = \mathbb{E}[D_{\text{stat}}(T_{\text{fake}})] - \mathbb{E}[D_{\text{stat}}(T_{\text{real}})] + \lambda\, GP_{\text{stat}} \tag{33}$$

$$\mathcal{L}_{D_{\text{phys}}} = \mathbb{E}[D_{\text{phys}}(R_{\text{fake}})] - \mathbb{E}[D_{\text{phys}}(R_{\text{real}})] + \lambda\, GP_{\text{phys}} \tag{34}$$

The Wasserstein distance estimates used to monitor training are:

$$W_{\text{stat}} \approx \mathbb{E}[D_{\text{stat}}(\omega_{\text{real}})] - \mathbb{E}[D_{\text{stat}}(\omega_{\text{fake}})] \tag{35}$$

$$W_{\text{phys}} \approx \mathbb{E}[D_{\text{phys}}(R_{\text{real}})] - \mathbb{E}[D_{\text{phys}}(R_{\text{fake}})] \tag{36}$$

The generator is trained to minimize:

$$\mathcal{L}_G = -\mathbb{E}[D_{\text{stat}}(\omega_{\text{fake}})] - \mathbb{E}[D_{\text{phys}}(R_{\text{fake}})] \tag{37}$$

with gradients computed via the chain rule through both discriminator paths:

$$\frac{\partial \mathcal{L}_G}{\partial \theta_G} = \frac{\partial \mathcal{L}}{\partial \omega_{\text{fake}}} \cdot \frac{\partial \omega_{\text{fake}}}{\partial \theta_G} + \frac{\partial \mathcal{L}}{\partial R_{\text{fake}}} \cdot \frac{\partial R_{\text{fake}}}{\partial \omega_{\text{fake}}} \cdot \frac{\partial \omega_{\text{fake}}}{\partial \theta_G} \tag{38}$$

2.8 Autoregressive Rollout

The generator produces trajectories autoregressively. Given an initial condition ω^0, each subsequent frame is obtained by feeding the previous prediction back into the generator:

$$\omega^{t+1} = G(\omega^t, z), \quad z \sim \mathcal{N}(0, I) \tag{39}$$

A full trajectory of length T is therefore:

$$\mathcal{T} = \{\omega^0,\, G(\omega^0, z),\, G^2(\omega^0, z),\, \ldots,\, G^T(\omega^0, z)\} \tag{40}$$

This is equivalent to approximating the continuous-time operator $\partial_t \omega = F(\omega)$ by a discrete map $\omega^{t+1} \approx \omega^t + h(\omega^t)$, where h is learned by the FNO. The stochastic latent vector z is resampled at each step, allowing the model to represent the distributional uncertainty of turbulent trajectories rather than collapsing to a single deterministic path.

3 Methodology

3.1 Dataset

The dataset is sourced from the `scaomath/navier-stokes-dataset` repository on Hugging Face [**?**]. It consists of isotropic turbulence trajectories modeled by the 2D incompressible Navier-Stokes equations, generated using a pseudo-spectral solver with a 4th-order Runge-Kutta scheme for the convection term. Different initial conditions with varying peak wavenumbers of $\mathcal{O}(1)$ are included, all converging to the Kolmogorov inverse energy cascade. The dataset contains 4 different fields including vorticity, differential, stream and residue including 1,152 trajectories of 100 timesteps each on a 64×64 spatial grid each, yielding a tensor of shape $(4, 1152, 100, 64, 64)$ for the implementation we only used the vorticty field represented by $\omega(x, t) = \nabla \times u$.

3.2 Preprocessing

The vorticity field is first normalized using a min–max scaling to the range [0,1]. It is then standardized by subtracting the global mean and dividing by the global standard deviation computed over the entire dataset. Sliding windows of length $T_{seq} = 10$ are constructed over each trajectory, producing input–output pairs (ω_n, ω_{n+1}) for autoregressive training. This results in a total of 103,680 training windows. An 80/20 train–validation split is applied at the window level.

3.3 Implementation

The model is implemented in PyTorch and trained on a CUDA-enabled GPU. The architecture follows a dual-discriminator Wasserstein GAN with Gradient Penalty (WGAN-GP) framework, where the generator is a Fourier Neural Operator.

The generator G_θ receives the current vorticity field ω_n alongside a stochastic latent vector $z \sim \mathcal{N}(0, I)$ and produces the next timestep ω_{n+1}. Trajectories are generated autoregressively, feeding each predicted frame as input to the next step.

Two specialized discriminators are maintained independently throughout training, each with its own optimizer, learning rate scheduler, and Wasserstein distance estimate. The statistical discriminator D_{stat} receives complete generated trajectories and judges their distributional realism. The physical discriminator D_{phys} receives the Navier-Stokes residual field computed between consecutive frames and judges physical consistency independently of statistical appearance.

The Navier-Stokes residual is computed spectrally at each training step, with viscosity ν and forcing f sampled stochastically to reflect the variability observed in the dataset. Both discriminators are updated $n_{critic} = 2$ times per generator step. Gradient penalty is applied independently to each discriminator over interpolated inputs in their respective input spaces.

Training progress is monitored through five validation metrics computed at the end of each epoch over the held-out validation set: mean squared error against ground truth, Navier-Stokes residual magnitude, kinetic energy, enstrophy, and spectral correlation. Two independent Wasserstein distance estimates — one per discriminator — serve as the primary indicators of adversarial equilibrium.

The next table presents the hyperparameters used in the training:

Table 1: Training and model hyperparameters

Parameter	**Value**
Grid size (H, W)	64×64
Sequence length T	10
Epochs	30
Batch size	8
Fourier modes	12
Hidden channels	32
FNO layers	4
Critic steps n_{critic}	2
Latent dimension z	8
Time step Δt	0.05
Gradient penalty λ	10.0

Also a baseline MSE-FNO was trained to contrast step evolution in the context of the spurious attractor, as discussed later in the text.

4 Results

4.1 Training Metrics

The training procedure employed a dual-discriminator Wasserstein GAN with gradient penalty (WGAN-GP) framework, optimizing a Fourier Neural Operator (FNO) generator against two specialized critics: `D_stat`, which operates directly on vorticity trajectories, and `D_phys`, which operates on the spatial Navier–Stokes residual field. Both discriminators were trained with a learning rate of 10^{-4}, while the generator learning rate was halved to 5×10^{-5} after epoch 9 via a `ReduceLROnPlateau` scheduler conditioned on validation MSE.

4.1.1 Wasserstein Distances and Adversarial Losses

The Wasserstein distances for both critics increased monotonically during the first ten epochs and subsequently plateaued, indicating that the discriminators reached a stable equilibrium with the generator. Specifically, W_{stat} stabilized around 111 and W_{phys} around 1,320. This is consistent with standard WGAN dynamics: the critic score reflects distributional distance rather than classification confidence, and a plateau indicates that the generator produces samples whose distribution is consistently separated from the real data manifold by a stable margin.

The generator loss $\mathcal{L}_G$ grew steadily throughout training, reaching approximately 2,377 by epoch 30. This is the expected behavior in WGAN-GP training: as the discriminators improve their ability to distinguish real from generated samples, the adversarial signal pushes the generator loss upward. The discriminator losses $\mathcal{L}_{D_{\text{stat}}}$ and $\mathcal{L}_{D_{\text{phys}}}$ remained negative throughout training, reflecting the Wasserstein objective in which critics are rewarded for assigning higher scores to real samples and lower scores to generated ones.

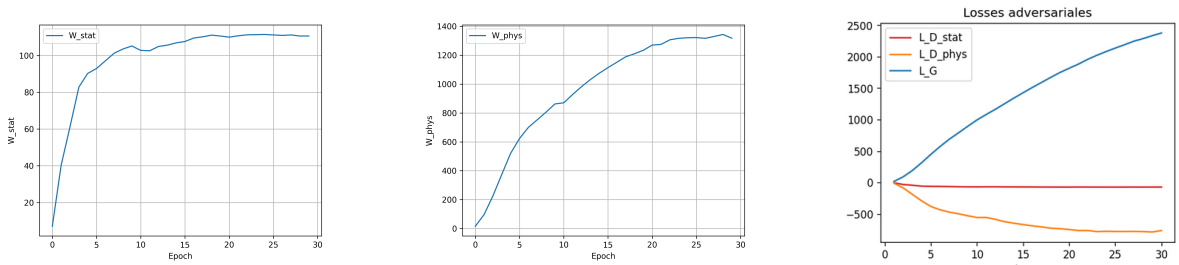

Figure 1: Wasserstein distances W_{stat} (left) and W_{phys} (center), and adversarial losses $\mathcal{L}_{D_{\text{stat}}}$, $\mathcal{L}_{D_{\text{phys}}}$, and $\mathcal{L}_G$ (right) over 30 epochs.

4.1.2 Validation MSE

The validation MSE decreased sharply from 0.528 at epoch 0 to approximately 0.248 by epoch 1, and continued decreasing gradually throughout training, reaching a best value of approximately 0.202 at epoch 27. Mild oscillations are present in later epochs, attributable to the stochastic nature of the adversarial objective and the learning rate reduction applied after epoch 9.

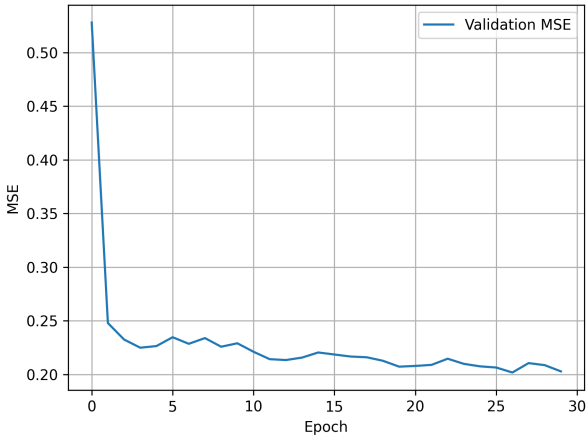

Figure 2: Validation MSE across epochs. The learning rate reduction at epoch 9 contributes to the accelerated convergence observed in subsequent epochs.

4.1.3 Physical Conservation Metrics

The ratio of generated to real energy and enstrophy fluctuates around 1.0 throughout training, indicating that the generator broadly preserves these integral invariants of the 2D Navier–Stokes equations. However, the oscillatory behavior—particularly in enstrophy, which exhibits excursions between 0.78 and 1.16—suggests that the physics discriminator provides a regularizing signal rather than enforcing strict conservation. The enstrophy ratio converges toward 1.0 in later epochs, consistent with the plateauing of W_{phys}.

The Navier–Stokes residual on the validation set increased from approximately 13 at epoch 0 to values oscillating between 17 and 20 throughout the remainder of training. This trend reflects the absence of explicit residual supervision: the adversarial objective encourages statistically realistic outputs but does not directly penalize equation violations. The residual therefore remains noisy and does not monotonically decrease, which is expected behavior in a zero-shot autoregressive generation setting where physical consistency is promoted implicitly through the physics discriminator rather than through a dedicated PDE loss term.

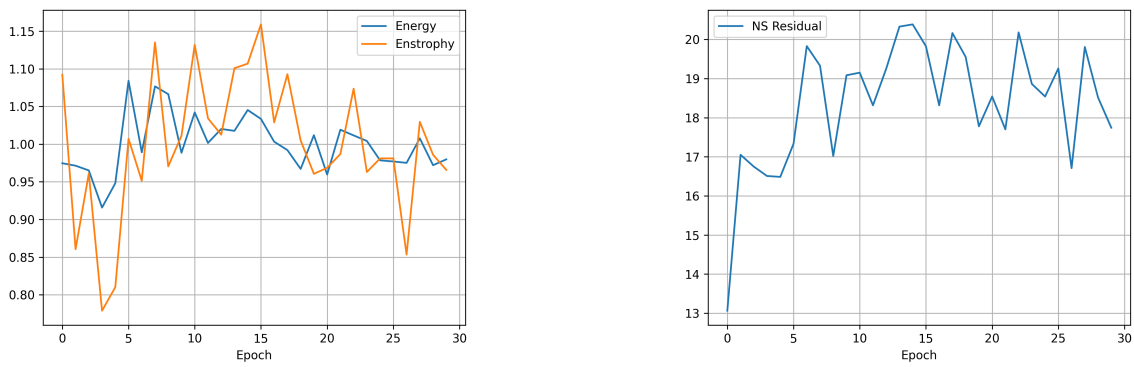

Figure 3: Left: ratio of generated to real energy and enstrophy over training. Values near 1.0 indicate physical consistency with the ground truth. Right: Navier–Stokes residual on the validation set. Persistent noise reflects the absence of explicit PDE supervision during training.

4.1.4 Spectral Diagnostics at Epoch 30

The spectral diagnostics at epoch 30 evaluate the quality of the generated turbulent fields in Fourier space, assessing whether the generator reproduces the correct distribution of energy across scales, the correct inter-scale energy transfer, and phase coherence with the ground-truth fields.

Enstrophy spectrum. The enstrophy spectrum $Z(k)$ of the generated fields closely follows the real spectrum across the resolved wavenumber range $k \in [1, 40]$. Both spectra peak near $k \approx 3$–4 and exhibit a plateau at intermediate wavenumbers consistent with the theoretical k^{-1} scaling for 2D turbulence in the enstrophy cascade regime. At high wavenumbers ($k > 10$), the generated spectrum exhibits slightly more variability but remains within one order of magnitude of the real spectrum.

Palinstrophy spectrum. The palinstrophy spectrum $P(k)$, which measures the squared gradient of vorticity and is sensitive to small-scale structures, is well reproduced by the generator across the full wavenumber range. Both spectra increase monotonically with k and lie well above the k^{+1} reference scaling, indicating that palinstrophy is concentrated at small scales as expected in 2D turbulence. The close agreement suggests that the generator captures fine-scale vorticity gradients accurately, despite these not being explicitly penalized during training.

Enstrophy transfer spectrum. The enstrophy transfer spectrum $T(k)$ characterizes the nonlinear inter-scale enstrophy flux. Both the real and generated spectra exhibit the same qualitative structure: positive transfer at large scales ($k < 4$), a pronounced negative peak near $k \approx 4$–5, and near-zero transfer at small scales. The generator reproduces the location and approximate depth of the negative peak, though it slightly overestimates the magnitude. This confirms that the adversarial training promotes correct nonlinear cascade dynamics without explicit transfer spectrum supervision.

Spectral correlation. The spectral correlation $C(k)$ measures phase coherence between generated and real fields in Fourier space, with $C(k) = 1$ indicating perfect agreement and $C(k) = 0$ indicating no coherence. At large scales ($k < 4$), the correlation is near or above 1.0, indicating that the generator faithfully reproduces dominant large-scale structures. The correlation decreases progressively at intermediate and small scales, dropping below 0.5 near $k \approx 8$ and approaching zero for $k > 15$. This decay is expected for an autoregressive model without explicit phase supervision. The final mean spectral correlation of approximately 0.30 across all wavenumbers indicates meaningful but limited phase coherence, consistent with the model learning the correct distribution of turbulent structures rather than predicting individual deterministic realizations.

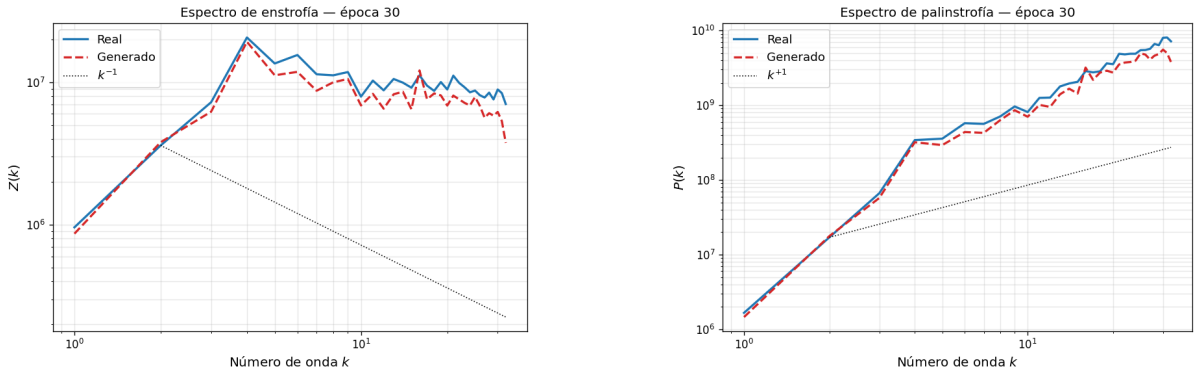

Figure 4: Left: enstrophy spectrum $Z(k)$ at epoch 30. The generated spectrum follows the real spectrum closely, with both consistent with the theoretical k^{-1} scaling at intermediate wavenumbers. Right: palinstrophy spectrum $P(k)$ at epoch 30, showing good agreement across all scales.

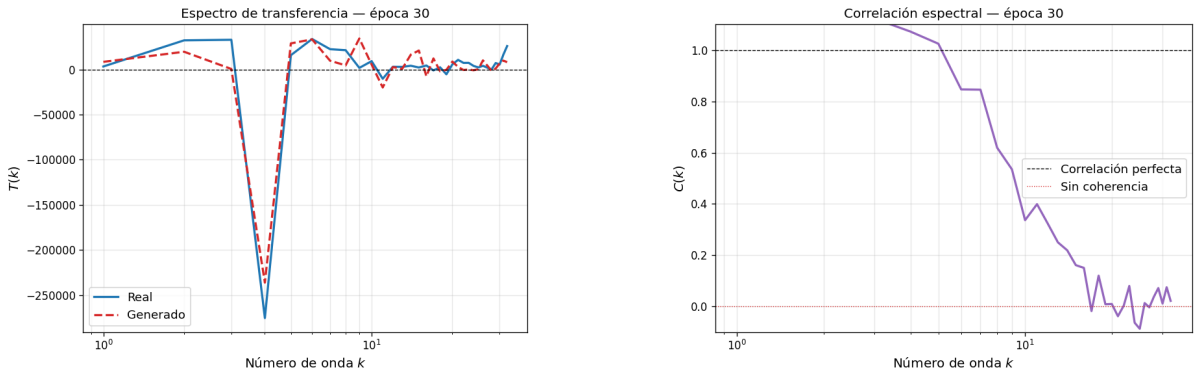

Figure 5: Left: enstrophy transfer spectrum $T(k)$ at epoch 30. The generator reproduces the characteristic cascade structure of 2D turbulence, including the negative peak near $k \approx 4$–5. Right: spectral correlation $C(k)$ at epoch 30. High correlation at large scales reflects accurate large-scale structure reproduction; the decay at small scales is consistent with the stochastic nature of turbulence generation.

These results confirm that adversarial training with physics-informed discriminators promotes statistically consistent turbulent structures without enforcing them through explicit equation constraints, and that the generator successfully approximates the spectral properties of the

ground-truth 2D turbulent fields across the resolved range of scales.

4.2 Super-Resolution

A fundamental property of Fourier Neural Operators is their formulation directly on function spaces, rendering the learned mapping independent of the discretization grid used during training. To validate this in the context of turbulent vorticity dynamics, a zero-shot super-resolution rollout was performed: the pretrained generator was applied autoregressively to an initial condition upsampled from 64×64 to 256×256 resolution, $\omega^0_{\mathrm{HR}} \in \mathbb{R}^{256\times256} = \mathrm{Interp}(\omega^0_{\mathrm{LR}})$, demonstrating the model's inherent capability to operate on finer grids without retraining.

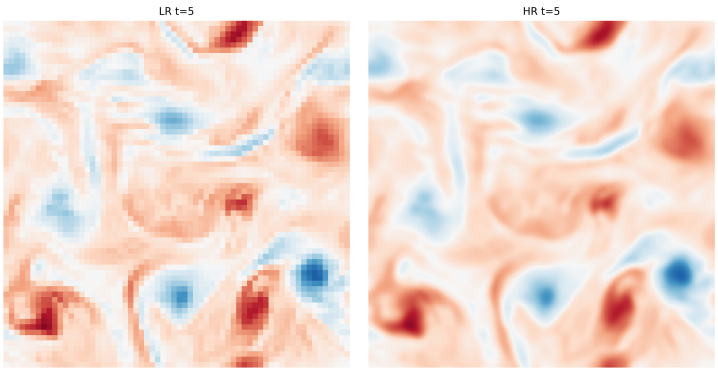

Figure 6: Super-Resolution Example.

4.3 Domain Decomposition

To assess the spatial generalization of the model, a domain decomposition strategy was applied, partitioning the full 64×64 vorticity domain into overlapping subdomains that are processed independently and subsequently reassembled. The physical diagnostic metrics confirm that the model preserves energy and enstrophy distributions across subdomain boundaries, and that the reconstructed global fields maintain physical consistency. This result suggests that the FNO generator has learned spatially local features that transfer across domain partitions without requiring global context, a desirable property for scalable simulation surrogates.

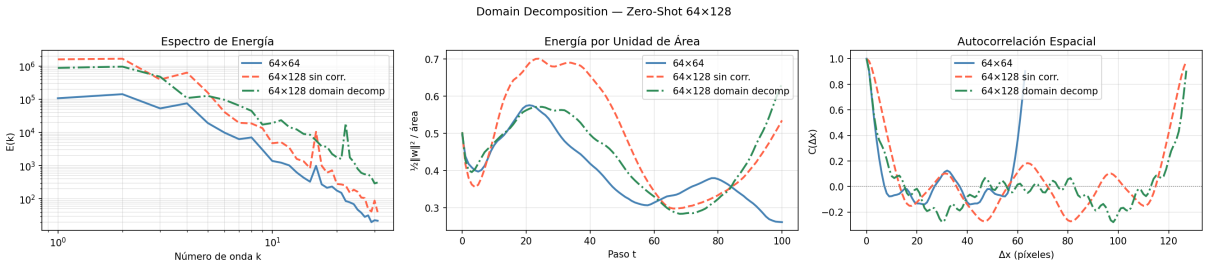

Figure 7: Analysis of Physical Metrics.

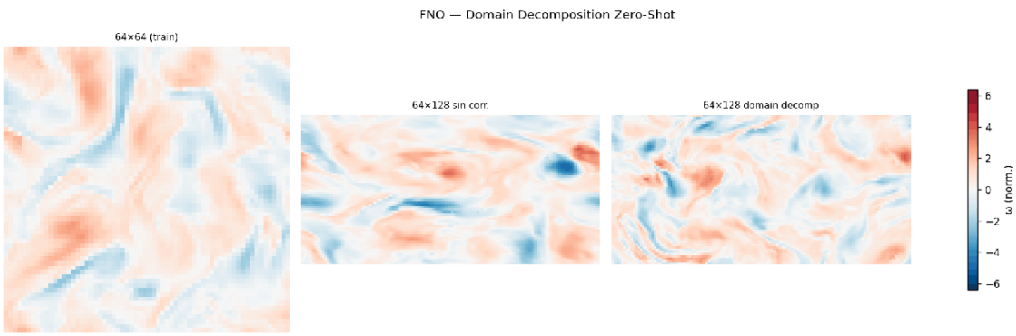

Figure 8: Domain Decomposition Results.

4.4 Inverse Boundary Method

The IBM zero-shot experiment evaluates the ability of the pretrained FNO generator to simulate vorticity dynamics in the presence of solid obstacles without any fine-tuning. Three obstacle geometries were tested: a centered circle of radius $r = 0.08$, an off-center circle of radius $r = 0.12$, and a centered rectangle. In each case, the immersed boundary condition $\psi = 0$ inside the obstacle was enforced at every autoregressive step via iterative volume penalization applied to the stream function recovered from the predicted vorticity field through a spectral Poisson solver. The algorithmic details of the immersed boundary implementation are provided in the Mathematical Foundations section.

The enstrophy evolution reveals that the penalization significantly alters the vorticity field produced by the surrogate model. This is primarily due to the reflection and scattering of the stream function induced by the obstacle, which modifies the local vorticity gradients and generates secondary flow structures

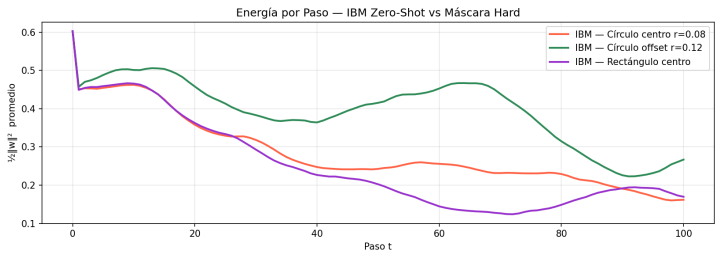

Figure 9: IBM Energy Analysis.

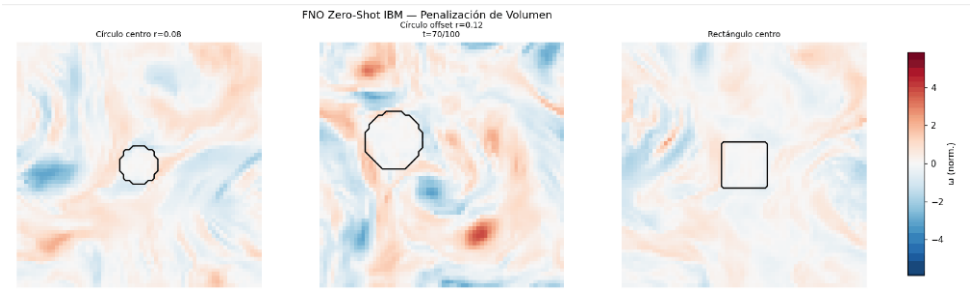

Figure 10: IBM Snapshot.

4.5 FID Distance

4.5.1 Baseline Comparison

The quality of the generated trajectories was assessed using the Fréchet Distance (FD) computed in the feature space of the two trained discriminators, following the methodology of the Fréchet Inception Distance but substituting the Inception network with domain-specific critics tailored to turbulent vorticity fields. Features from `D_stat` and `D_phys` were concatenated to form a joint representation capturing both statistical and physical characteristics of each trajectory.

Formally, let $\boldsymbol{\mu}_r, \boldsymbol{\Sigma}_r$ and $\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g$ denote the mean vectors and covariance matrices of the concatenated features computed from real and generated trajectories, respectively. The Fréchet Distance is then defined as:

$$\mathrm{FD} = \left\|\boldsymbol{\mu}_r - \boldsymbol{\mu}_g\right\|_2^2 + \mathrm{Tr}\left(\boldsymbol{\Sigma}_r + \boldsymbol{\Sigma}_g - 2\left(\boldsymbol{\Sigma}_r\boldsymbol{\Sigma}_g\right)^{1/2}\right) \tag{41}$$

The FD was computed over 10 independent random subsets of 100 validation samples and reported as mean $\pm$ standard deviation.

As shown in Table, the reference value $\mathrm{FD}(\mathrm{GT}, \mathrm{GT}) = 0.0023 \pm 0.0005$ establishes the natural variability floor of the metric under sampling noise. The GAN-based generator (Gen) achieves $\mathrm{FD} = 0.2507 \pm 0.0159$, while the MSE-trained baseline obtains $\mathrm{FD} = 0.0783 \pm 0.0159$. The lower FD of the baseline is explained by its training objective: direct MSE supervision with a larger model capacity (`hidden_ch` $= 64$ versus 32 for Gen) encourages the generator to produce trajectories that are metrically close to the ground truth in an ℓ_2 sense, which is directly reflected in the discriminator feature space. The GAN generator, by contrast, optimizes a distributional objective and does not minimize pointwise reconstruction error, leading to a higher FD despite producing trajectories with richer turbulent structure and better spectral characteristics.

Visual inspection of the feature embeddings reveals that the Baseline cluster overlaps substantially with the Ground Truth distribution, whereas the Gen cluster is shifted, indicating a systematic distributional offset. Notably, this offset is not fixed: with extended training, the generator's feature distribution gradually drifts toward the real manifold, suggesting that the reported FD for the GAN model could be further reduced by additional epochs or improved hyperparameter tuning.

Table 2: Fréchet Distance (FD) comparison

Comparison	**FD (mean ± std)**
GT vs GT	0.0023 ± 0.0005
GT vs Gen	0.2507 ± 0.0159
GT vs Baseline	0.0783 ± 0.0159

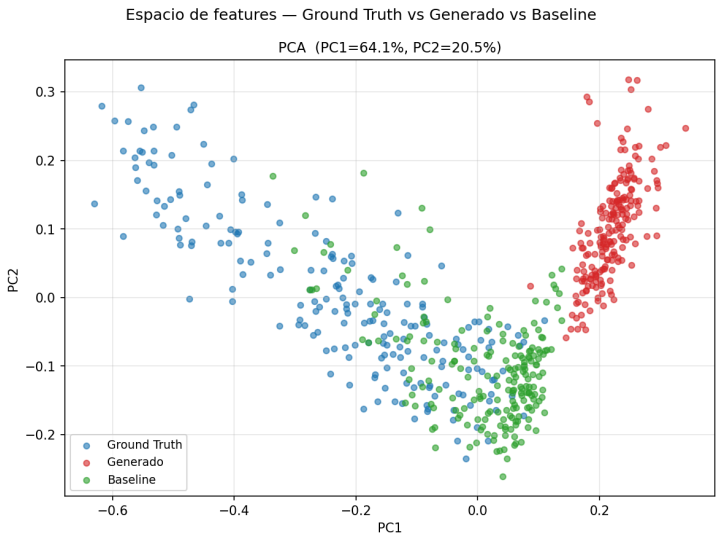

Figure 11: PCA of feature space.

4.5.2 Fourier Spurious Attractor

A notable failure mode was identified during autoregressive rollouts extended beyond the training horizon. When the generator is unrolled for more than approximately 75 steps — well past the training sequence length of 10 — the predicted vorticity fields begin to exhibit a spurious diagonal wavenumber pattern accompanied by vertical striping artifacts (Fig. **??**).

This behavior is consistent with the emergence of a non-physical attractor in Fourier space. In the autoregressive setting, each prediction is fed back as input, causing the trajectory to explore regions of the latent space and energy spectrum that lie far from the training distribution. Under these conditions, the Fourier Neural Operator tends to revert to the dominant linear dynamics embedded in its learned weights — primarily those associated with horizontal wavenumber bands. The observed spurious diagonal and striping patterns correspond to residual Fourier components that were present during training but become unstable and self-amplifying when iterated beyond the intended temporal window. This highlights a known vulnerability of autoregressive FNO rollouts: the accumulation of spectral bias errors leads to a gradual drift away from the physical manifold of the Navier–Stokes equations.

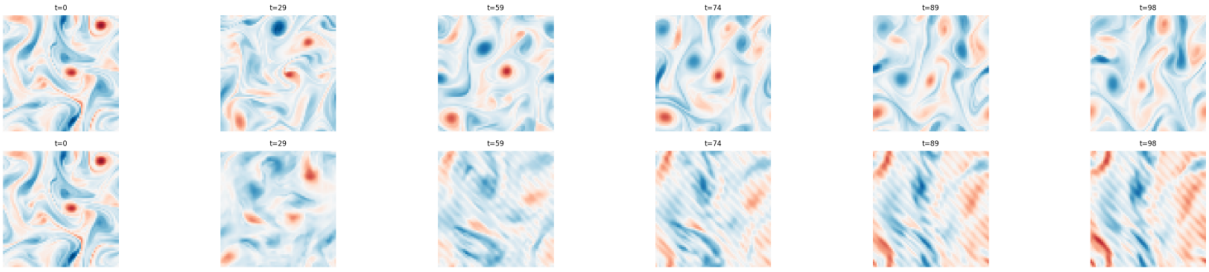

Figure 12: Appearance of spurious attractor over time.

5 Conclusions

5.1 Future Work

References

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[2] M. A. Rahman, M. A. Florez, A. Anandkumar, Z. E. Ross, K. Azizzadenesheli, *Generative Adversarial Neural Operators*, arXiv:2205.03017, 2022. https://arxiv.org/abs/2205.03017